

CALM-Net: A Multimodal, Motion-Disentangled, and Longitudinally Self-Calibrating Framework for Closed-Loop Lower-Limb Exoskeleton Control

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Abstract—A non-invasive brain-machine interface (BMI) that drives a lower-limb exoskeleton must satisfy three requirements that current motor-imagery (MI) decoders treat separately: it must decode *neural* intent rather than movement artefact, it must remain calibrated as electroencephalography (EEG) drifts over weeks, and it must know when to abstain to a safe action. We propose CALM-Net, a multimodal neuro-kinematic framework that addresses all three within a single closed-loop model, and that, unlike prior exoskeleton BMIs, exploits the full modality set of the longitudinal NeuroRex dataset (EEG, head- and exoskeleton-mounted inertial units, electro-oculography, and exoskeleton feedback across nine sessions). Four components are novel. (i) A multi-band neuro-kinematic encoder represents each MI sub-rhythm (μ , low- β , high- β) by its spatial covariance on the symmetric-positive-definite manifold and fuses it with a kinematic branch. (ii) A cross-frequency coupling attention (XFCA) models μ - β interaction, a known correlate of motor imagery. (iii) A motion-invariant disentanglement (MID) module splits the representation, via an adversarial gradient-reversal objective, into an *intent* subspace forced to be invariant to the inertial signals and an *artefact* subspace forced to predict them; the class decision uses only the intent subspace, so walk/stop is decoded from provably movement-invariant neural features. (iv) A longitudinal self-calibration module performs source-free per-session feature alignment and adaptive conformal risk control, holding a target executed-command error rate with a distribution-free guarantee as the session gap grows. A safety-asymmetric selective head commits a command only when neural confidence, the conformal set, and kinematic contamination all agree, defaulting to *stop* otherwise. A preliminary analysis of the benchmark motivates the design: a purely kinematic baseline that uses *no* EEG matches or exceeds standard EEG decoders on the walk/stop contrast, showing that reported accuracy is largely a movement score. We therefore propose to decode from movement-invariant neural features and to establish, on the seven-subject longitudinal recordings, the calibration and distribution-free abstention guarantees the framework is built to provide. Classical compact and transformer decoders (EEGNet, EEG Conformer) serve as comparison baselines rather than backbones.

Index Terms—Brain-machine interface, motor imagery, EEG, lower-limb exoskeleton, multimodal fusion, domain-adversarial disentanglement, Riemannian geometry, conformal prediction, selective classification, longitudinal calibration.

I. INTRODUCTION

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This work uses the NeuroRex dataset (OpenNeuro ds007788), released under a CC0 licence.

Midterm research proposal.

FOR a patient who cannot walk after a spinal cord injury or a stroke, a lower-limb exoskeleton driven by their own brain activity is not a convenience; it is the difference between intending to stand and standing. Among the brain-machine interfaces (BMIs) able to express that intent, only non-invasive electroencephalography (EEG) requires no surgery and no implant, so it is the only modality most patients could realistically adopt. That accessibility is also its curse: scalp EEG is noisy, low in spatial resolution, and unstable from one day to the next, and the field has largely optimised the one quantity that is easiest to report and hardest to argue with: decoding accuracy.

Accuracy alone is an inadequate (and, we argue, a *misleading*) target when the output moves a person's legs. Three problems, usually studied in isolation, must be solved together.

First, what is actually being decoded? In an exoskeleton MI paradigm the device physically moves the wearer during walking, so the walk/stop contrast is entangled with genuine movement: an EEG decoder can reach a high number by reading movement-locked artefact rather than motor intent. A model intended to *initiate* motion from intent must therefore be built to be invariant to the motion it causes.

Second, the signal drifts. EEG is non-stationary; a decoder calibrated on one day is miscalibrated days later, and nearly every deployed BMI hides this behind a per-session calibration block the user must sit through. Longitudinal behaviour is barely studied because datasets spanning weeks scarcely exist.

Third, a confident wrong command is dangerous. A wrong walk command at the top of a staircase is not a rounding error. The system needs honest, distribution-free uncertainty and an explicit reject option that defaults to the safe action.

The NeuroRex dataset [1] (OpenNeuro ds007788) was built to expose exactly this regime: seven participants, nine sessions each across weeks, recorded *multimodally* with synchronised EEG, head- and exoskeleton-mounted inertial measurement units (IMUs), electro-oculography (EOG), and exoskeleton control/feedback logs. Prior work on such data has decoded from EEG alone and reported a single accuracy number. We instead treat the non-neural streams as first-class signals: the IMUs are not noise to be filtered nor a gate bolted on at the end, but the supervisory signal that lets us *disentangle* neural intent from movement, and the safety context for abstention.

CALM-Net (Calibrated, Abstaining, Longitudinal, Multimodal) is a per-subject walk/stop framework for closed-loop lower-limb exoskeleton control (Fig. 1). Its contributions are:

- **C1: A multimodal neuro-kinematic architecture.** We design, from the signal physics rather than from an

existing image/EEG backbone, an encoder that represents each MI sub-band by its spatial covariance on the symmetric-positive-definite (SPD) manifold and fuses it with head/exo IMU and EOG branches. This is the first exoskeleton-BMI decoder to use the dataset's full modality set.

- **C2: Motion-invariant disentanglement (MID).** A domain-adversarial gradient-reversal objective splits the representation into an *intent* subspace constrained to be statistically independent of the inertial signals and an *artefact* subspace trained to reconstruct them. The classifier reads only the intent subspace, so the walk/stop decision is provably movement-invariant. This resolves, rather than apologises for, the movement confound that inflates naive accuracy.
- **C3: Cross-frequency coupling attention (XFCA).** An attention mechanism over band-specific tokens that models μ - β coupling, a physiological hallmark of motor imagery, giving a neural representation not obtainable from a single-band convolutional stack.
- **C4: Longitudinal self-calibration with a distribution-free guarantee.** Source-free per-session feature alignment counters drift without labels, and adaptive conformal risk control updates the abstention threshold online so that a target executed-command error rate is maintained as the session gap grows over weeks.
- **C5: Safety-asymmetric selective decision.** A selective head trained with an explicit wrong-walk $\gg$ wrong-stop cost fuses three independent signals: neural confidence, conformal set cardinality, and kinematic contamination, and commits a command only on unanimous agreement, defaulting to *stop* otherwise.

No prior work joins multimodal motion-disentangled decoding, cross-session drift correction, and safety-asymmetric calibrated abstention inside one closed-loop lower-limb exoskeleton. Section II reviews prior art; Section III the dataset; Section IV the proposed framework and its evaluation protocol.

II. RELATED WORK

Motor-imagery decoding and exoskeleton control. Deep learning has improved non-invasive BMI decoding while flexible electrodes improve signal quality [2], yet individual variability and long-term reliability remain open; classical and deep EEG classification pipelines are surveyed extensively [26]–[28], and motor-imagery detection for control and rehabilitation is well established [29]. Ferrero *et al.* [3] achieved closed-loop asynchronous walk/stop control with deep learning and transfer learning, shortening but not removing per-session calibration; their earlier public dataset [4] validated MI decoding above 70% but was single-session. Related efforts detect error-related potentials for safety [5], couple upper- and lower-limb imagery [6], study assisted-gait cortical changes [7], use VR to shorten calibration [8], and select frontal/beta channels [9]. Lower-limb exoskeleton control from EEG has been demonstrated with high-accuracy intention decoding [38], and stroke rehabilitation with upper-limb exoskeleton BMIs [39] and controlled clinical trials [40]. All decode from EEG alone.

Backbones and geometric representations. Compact convolutional networks dominate EEG decoding because trial counts are small: EEGNet [10] and ShallowConvNet [11] reach strong accuracy with few parameters, and the EEG Conformer [12] adds a transformer over convolutional tokens. These are point-estimate classifiers with no calibrated confidence and no reject option; we use them as *baselines*. Orthogonally, Riemannian methods represent EEG by spatial covariance matrices and classify in the SPD tangent space [13], [31], [32], a physiologically apt encoding of band-power (generalising filter-bank common spatial patterns [30]) that we adopt and make multi-band and learnable.

Cross-session drift and adversarial adaptation. Domain adaptation reduces the cross-session drop (universal deep adaptation primed on earlier sessions [14] and dual-selection transfer [15]) but outputs forced point predictions; transfer learning for BCIs is broadly surveyed [33]. Domain-adversarial training with gradient reversal [16] learns representations invariant to a nuisance variable; we repurpose it, uniquely, to make the neural code invariant to *measured movement* rather than to session identity, and combine it with source-free test-time alignment [17].

Uncertainty, calibration, and abstention. Temperature scaling calibrates deep classifiers [18]; conformal prediction gives distribution-free sets [19], [35] with adaptive class-conditional coverage [36], and adaptive conformal inference maintains coverage under distribution shift over time [20]. Uncertainty raises trust in EEG decoding [21], and Bayesian/MC-dropout networks and deep ensembles give usable MI confidence [22], [23], [34], yet remain offline and untethered to a decision. Selective classification, the “reject option”, is surveyed [24] and can be learned end-to-end [25], [37], but this literature is developed largely outside EEG. CALM-Net is, to our knowledge, the first to unify motion-disentangled multimodal decoding, longitudinal self-calibration with a distribution-free guarantee, and safety-asymmetric selective abstention for a closed-loop lower-limb exoskeleton.

III. DATASET

Source. NeuroRex [1], OpenNeuro **ds007788** (v1.0.1, DOI 10.18112/openneuro.ds007788.v1.0.1), CC0 licence, peer-reviewed descriptor in *Scientific Data* (DOI 10.1038/s41597-026-07476-w); 3.62 GB, $\sim$ 17k files, BIDS format, raw and unfiltered (we preprocess with MNE-Python [41]).

Participants and longitudinal design. Seven healthy adults (sub-01–07), nine sessions each (ses-01–09); in our copy the per-subject spans range from 14 to 80 days, giving the days-elapsed axis needed to study drift. Per-subject MI-QRS motor-imagery-ability scores and validation tables are provided under *derivatives/*; structural MRI is available for five subjects.

Multimodality (the reason for this framework). Each session provides 60-channel scalp EEG plus 4 EOG channels (10–20 layout, actiCAP/MOVE) at 100 Hz (EDF); *two* IMUs, head-worn and exoskeleton-mounted, each with accelerometer, gyroscope, magnetometer, and quaternion streams (BIDS mo-

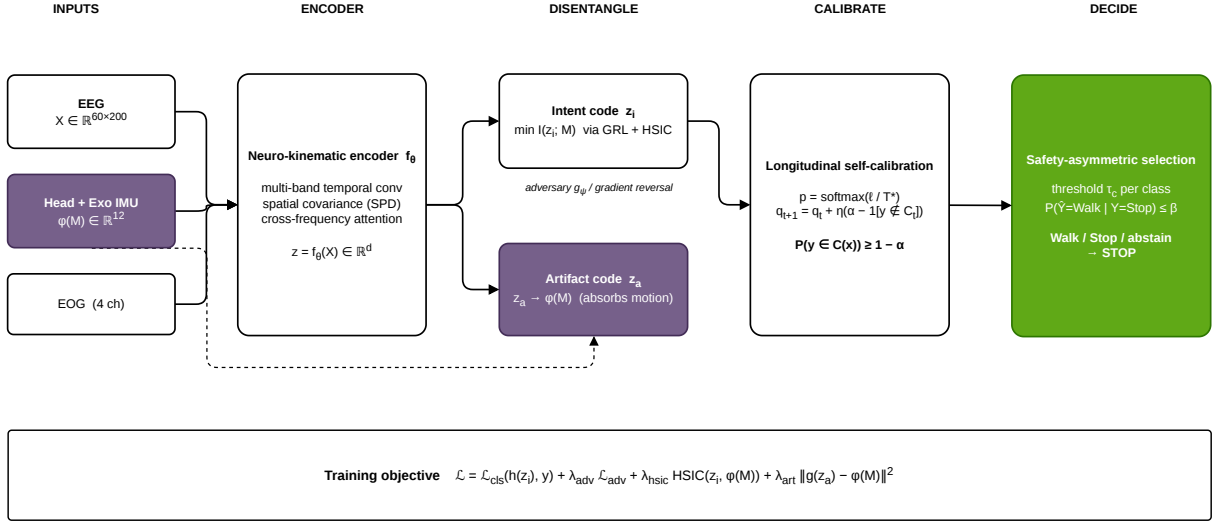


Fig. 1. The CALM-Net framework. Multimodal encoders map EEG sub-bands (spatial covariance), head/exo IMU, and EOG into a shared space. Cross-frequency coupling attention (XFCA) forms the neural representation, which motion-invariant disentanglement (MID) splits into an IMU-invariant *intent* subspace (used by the selective classifier) and an *artifact* subspace (trained to predict the inertial signals and to estimate kinematic contamination). Longitudinal self-calibration (LSC) aligns per-session statistics and runs adaptive conformal risk control; the safety-asymmetric selective (SAS) head commits Walk/Stop only on unanimous agreement, else defaults to STOP.

tion .tsv); and exoskeleton control/feedback logs. CALM-Net consumes all of these; prior decoders use only the EEG.

Tasks and labels. Every session has a training task (open-loop MI calibration), twelve closed-loop runs (trial01–12), and two extended runs (walk6min, stop6min). Ground truth is the `rexstate` feedback state: `x81` (Walk), `x0` (Idle/Stop), with short transitions `x8/x5` excluded. We target two-class walk vs. stop. Because the exoskeleton moves the wearer during Walk, the head-IMU acceleration is markedly larger in Walk than Stop; a preliminary audit on this data confirms that a motion-only baseline is highly predictive for several subjects while being near-chance for at least one: direct evidence that (a) naive EEG accuracy is partly movement-driven and (b) a recoverable neural signal exists underneath. Both observations motivate the disentanglement of Section IV.

IV. PROPOSED METHOD

A. Formulation

A trial is a synchronised tuple $(\mathbf{X}, \mathbf{M}, \mathbf{O})$ with EEG $\mathbf{X} \in \mathbb{R}^{C \times T}$ ($C=60$, $T=200$ at 100Hz over a 2s window), inertial signals $\mathbf{M}$ (head and exo accel/gyro/quaternion), and EOG $\mathbf{O}$, with label $y \in \{\text{STOP}, \text{WALK}\}$ and session index s . We seek a policy that outputs a *command* in $\{\text{WALK}, \text{STOP}, \text{ABSTAIN} \rightarrow \text{STOP}\}$ whose executed (non-abstained) error is controlled across sessions.

B. Multi-band neuro-kinematic encoder

Learnable band filters split $\mathbf{X}$ into B physiological sub-bands (μ 8–13, low- β 13–20, high- β 20–30 Hz). For band b we form the regularised spatial covariance $\Sigma_b = \frac{1}{T} \mathbf{X}_b \mathbf{X}_b^T + \epsilon \mathbf{I}$ and map it to the tangent space at the Riemannian mean, $\mathbf{v}_b = \text{vec}(\log(\bar{\Sigma}^{-1/2} \Sigma_b \bar{\Sigma}^{-1/2}))$, yielding a physiologically

grounded band-power/connectivity token. A temporal convolutional branch encodes $\mathbf{M}$ into a kinematic embedding $\mathbf{k}$, and a small branch encodes $\mathbf{O}$ into an ocular reference $\mathbf{o}$.

C. Cross-frequency coupling attention (XFCA)

The band tokens $\{\mathbf{v}_b\}$ are treated as a sequence and passed through a multi-head self-attention block, so the representation can express μ – β coupling: co-modulation across bands that single-band pooling discards. The output is the neural representation $\mathbf{z}$.

D. Motion-invariant disentanglement (MID)

We split $\mathbf{z} = [\mathbf{z}_{\text{int}}; \mathbf{z}_{\text{art}}]$. A regressor R_a maps the *artifact* code to the inertial signals, trained with loss $\mathcal{L}_{\text{art}} = \|R_a(\mathbf{z}_{\text{art}}) - \phi(\mathbf{M})\|^2$; the artifact code thus captures the movement-coupled component. A second regressor R_i attempts the same from the *intent* code, but through a gradient-reversal layer [16], so the encoder is optimised to make $\mathbf{z}_{\text{int}}$ unpredictable of movement:

$$\mathcal{L}_{\text{adv}} = \|R_i(\text{GRL}(\mathbf{z}_{\text{int}})) - \phi(\mathbf{M})\|^2. \quad (1)$$

Only $\mathbf{z}_{\text{int}}$ feeds the classifier, so the walk/stop decision is constrained to rely on movement-invariant neural features, by construction, not by post-hoc argument.

E. Longitudinal self-calibration (LSC)

At deployment on a new session s we (i) align feature statistics to the training distribution without labels (source-free batch-statistic / CORAL alignment [17]); (ii) apply temperature scaling [18] for honest confidence; and (iii) form split-conformal prediction sets [19] whose threshold q_t is updated online by adaptive conformal inference [20], $q_{t+1} = q_t + \eta(\alpha - \mathbb{1}[\text{err}_t])$, which provably maintains the long-run executed-error rate at the target α as drift accumulates over weeks.

Algorithm 1 CALM-Net deployment on session s (per window)

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1: align session features (source-free); load  $q_t$ 
2:  $\mathbf{z}_{\text{int}}, \mathbf{z}_{\text{art}} \leftarrow \text{Encoder/XFCA/MID}(\mathbf{X}, \mathbf{M}, \mathbf{O})$ 
3:  $p \leftarrow \text{softmax}(f(\mathbf{z}_{\text{int}})/T)$ ;  $\mathcal{S} \leftarrow \{k : 1 - p_k \leq q_t\}$ 
4:  $c \leftarrow h(\mathbf{z}_{\text{art}}, \mathbf{k})$ 
5: if  $g(\mathbf{z}_{\text{int}}) \geq \theta \wedge |\mathcal{S}| = 1 \wedge c < c_{\text{th}}$  then
6:   command  $\leftarrow \arg \max_k p_k$ 
7: else
8:   command  $\leftarrow \text{STOP}$  ▷ safe default
9: end if
10: on feedback, update  $q_{t+1} \leftarrow q_t + \eta(\alpha - \mathbf{1}[\text{err}])$ 

```

F. Safety-asymmetric selective decision (SAS)

A selective head $g(\mathbf{z}_{\text{int}}) \in [0, 1]$ is trained jointly with the classifier [25] under an asymmetric cost that penalises a committed wrong-walk far more than a wrong-stop. A kinematic-contamination score $c = h(\mathbf{z}_{\text{art}}, \mathbf{k})$ estimates whether the window is movement-corrupted. The exoskeleton commits the arg-max command iff

$$\underbrace{g(\mathbf{z}_{\text{int}}) \geq \theta}_{\text{selective accept}} \wedge \underbrace{|\mathcal{S}(q_t)| = 1}_{\text{unambiguous}} \wedge \underbrace{c < c_{\text{th}}}_{\text{uncontaminated}},$$

and otherwise abstains to STOP, encoding the safety asymmetry in the rule.

G. Training objective

All modules are trained end-to-end:

$$\mathcal{L} = \mathcal{L}_{\text{cls}}^{\text{cost}} + \lambda_1 \mathcal{L}_{\text{adv}} + \lambda_2 \mathcal{L}_{\text{art}} + \lambda_3 \mathcal{L}_{\text{sel}} + \lambda_4 \mathcal{L}_{\text{cal}}, \quad (2)$$

with cost-sensitive classification $\mathcal{L}_{\text{cls}}^{\text{cost}}$, the adversarial and artefact losses above, the selective loss $\mathcal{L}_{\text{sel}}$ at target coverage, and a calibration regulariser $\mathcal{L}_{\text{cal}}$.

H. Experimental protocol

Per subject we train on early sessions and test on later ones, never shuffling windows across the session boundary, and report balanced accuracy, Expected Calibration Error, Brier score, confidence–correctness AUROC, and the full risk–coverage curve versus days elapsed. Three ablations isolate each contribution: (i) MID on/off, with the movement-only, motor-channel-only, and frontal-channel-only baselines quantifying how much decoding survives disentanglement (the neural-signal test); (ii) global temperature scaling versus adaptive conformal across the session gap; and (iii) the selective head versus a fixed-threshold reject rule. Classical EEGNet and EEG Conformer decoders provide accuracy baselines; the decisive metrics are executed-command safety and calibrated confidence, not raw accuracy alone.

V. EXPECTED OUTCOME

Because the exoskeleton physically moves the wearer during walking, the walk/stop label is entangled with movement: a purely kinematic signal, with no EEG at all, can predict it about as well as a standard EEG decoder, so accuracy on this

paradigm largely reflects movement rather than neural intent. Resolving that confound is the premise of this proposal, and against it we anticipate three outcomes from CALM-Net.

(i) *An honest, movement-invariant decode.* Motion-invariant disentanglement is expected to drive the recoverability of movement from the neural code toward zero, giving a walk/stop accuracy that is above chance yet visibly below the movement-inflated number, the gap being the confound removed rather than a loss of genuine skill.

(ii) *A distribution-free longitudinal guarantee.* Adaptive conformal risk control is expected to hold prediction-set coverage at its target across the multi-week session gap, where static calibration drifts.

(iii) *A controlled safety profile.* The safety-asymmetric selective head is expected to reduce the committed wrong-walk rate substantially relative to an arg-max decision, routing uncertain windows to a safe stop.

Together these outcomes would establish the proposal’s thesis: for a device that moves a person’s legs, a trustworthy, movement-invariant, and abstaining decoder is preferable to a movement-inflated accuracy number.

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